Cahier des Charges – Day of Integration Platform

Project specification for a web app that runs the team games, scoring, rotation and clue system of the Day of Integration.

1. Context and Objectives

The Day of Integration brings together students from two years (A1 and A2) and several classes. Students take part either in **teams** (team games and board games, all day) or as **solo players** (board games, for fun and outside the reward system). The platform must:

- Display live team standings, compositions and current games to all participants.
- Let admins (game masters) control their games: start, timers, results, and rotation.
- Automatically build balanced teams from the participation form.
- Run a **clue system**: each win gives the team one clue, hidden until the final round.

2. Scope

In scope: public landing page, solo players page, admin dashboard, team generation, game rotation, scoring, clue management, final reveal.

Out of scope (for now): the 4 board games' detailed rules and scoring (to be added later), payments, user accounts for participants.

3. Actors

Actor	Description
Team member / Visitor	Student who views the public pages and their own team page (current games, score, standing). No login required.
Game Master	Staff member responsible for one or more games. Starts and ends them, designates the winner and gives the clue. Must log in.

Actor	Description
Admin (Organizer)	Imports form data, generates teams, manages games, assignments and the final reveal. Can also be game master of one or several games. Must log in.

4. Games

4.1 Team games (9)

1. Code Names
2. Toxic Waste
3. Water Transfer
4. Dancing Mannequins
5. Sac Jump
6. Spotting
7. Flip 7
8. Pull the Rope
9. Spot the Bugs

4.2 Board games (4)

Names and rules to be defined later. Board games are played by teams as part of their day and by solo players. The data model must support them as a second game category (`BOARD`) so they can be added without changes to the schema.

5. Input Data: Participation Form

Field	Type	Values
Full name	text	free
Year	enum	A1, A2
Class name	enum	4 classes per year

Field	Type	Values
Gender	enum	Male, Female
Participation mode	enum	Team or Solo (a field of the form)

Data is imported by an admin (CSV upload or manual entry) and stored in the database.

Team size is fixed at 6 players. The number of teams is not fixed: it equals the number of team registrations ÷ 6, computed once the form closes. Every rule below uses this derived number of teams (T).

6. Functional Requirements

6.1 Public side (user)

FR-U1 – Landing page: teams

- Display all teams (T = team players ÷ 6) with name, score (number of clues collected) and rank (live leaderboard).
- Display each team's composition (members with year and class).
- Highlight the top teams.
- Show the game each team is currently playing (or "waiting / transit") and the status of that game (not started, running, finished).
- Auto-refresh without manual reload.

FR-U2 – Solo players page

- List solo players by name.
- Show the games they are collectively in (the board games), and their current status. Solo players have no points, no clues and no rank: their games are for fun and outside the reward system, and only teams can win the final prize.

FR-U3 – Clues

- Public pages show only the **number** of clues a team has collected, never their content, until the organizer triggers the final reveal.

FR-U4 – Team page (no login)

- A participant opens their team page from the landing page by picking their team, or through a QR code or short link per team.

- It shows the game the team should go to now, the next games in its rotation, and the status of each.
- It shows the team's current score (clues collected) and its rank in the league leaderboard.

6.2 Admin side

FR-A1 – Authentication. All staff (admins and game masters) must log in with credentials; participants never log in. Roles: SUPER_ADMIN (one person only), `ADMIN`, `GAME_MASTER`. A game master only sees and controls the games they are assigned to. An admin sees everything and can also be assigned as game master of one or several games, so one person can hold both roles.

FR-A2 – Dashboard. Shows the games the admin is responsible for, the teams currently in each game, game status, and the timer.

FR-A3 – Game control.

- **Start game:** one action starts the timer for all teams/members in that game at the same time (server-side start timestamp, so every device shows the same time).
- **End game:** one action stops the timer, records the result, and triggers the rotation.
- Designate the winning team, which receives the clue.
- Ability to correct a mistaken result (with an audit log).

FR-A4 – Clue assignment. At game end, the game master gives **one clue** to the winning team. The clue content is entered in the platform and stored hidden.

FR-A5 – Rotation. When a game ends, teams are moved to their next game following the rotation plan (see 7.2). An admin can override the assignment manually.

FR-A6 – Team generation (admin). Import the form data, run the balancing algorithm, preview the teams, regenerate, manually swap members, then lock the teams.

FR-A7 – Final reveal. Once all games are finished and every team completed its rotation, an admin unlocks the final round. Each team then sees its collected clues and gives its answer orally, all at the same time. The first correct answer wins the prize.

FR-A8 – Freeze games (super admin only). There is one super admin (`SUPER_ADMIN`), and only this person can freeze or unfreeze a game. A frozen game is taken out of the rotation pool; unfreezing puts it back. The rotation schedule is regenerated from the games currently in the pool.

7. Business Rules and Algorithms

7.1 Team balancing

Goal: teams of 6 students each (T = registered team players ÷ 6) that are as **diverse** as possible.

- Spread years (A1/A2) evenly across teams.
- Spread classes evenly: avoid several students of the same class in the same team.
- Spread genders evenly, ideally per class and per year, so a team does not get the same gender of the same class or year concentrated.
- Randomness is used to break ties, so results differ at each generation.

Proposed approach: random initial distribution, then iterative improvement (swap two students from different teams if the swap reduces a penalty score).

Penalty = sum over teams of: imbalance in year + imbalance in gender + number of pairs sharing the same class + number of pairs sharing the same (class, gender) combination.

The algorithm stops when the penalty reaches a minimum or after N iterations. It is fast for this size (dozens to a few hundred students).

7.2 Rotation between games

Each team must play every team game once (full rotation). The number of teams T is unknown until registration closes, so the schedule is generated from T and from the games currently in the rotation pool (see FR-A8). If T is below 9, the super admin freezes games out of the pool so the rotation fits the number of teams. If T is above 9, a team that has finished its rotation of 9 games stops playing team games and goes to the board games, where it can keep playing; its progress and clues are kept, and the clues are revealed only once every team is done (see section 12). The rotation is generated before the day as a schedule (round × team → game) such that:

- Each match is team vs team, a team is in only one match at a time, and the winner of the match gets the clue.
- Each team plays each game exactly once.
- When a game ends, the next-round assignment is derived from the schedule, with random order inside the constraints.

7.3 Scoring

- The score of a team is the number of clues it has collected. There are no separate points.
- A team earns one clue each time it wins a game, so the maximum is one clue per game played.
- The ranking is the clue count. Ties are broken in the final round, and if the tied teams give the same answer, one last game of luck (like a penalty shootout) decides (see section 12). Solo players have no score or rank: only teams can earn clues and win the final prize.

7.4 Clue rules

- One clue per game, given to the winning team only.
- The clue is stored server-side and **never sent to the client** until the reveal is unlocked.
- Number of clues per team is public, content is private.

8. Technical Architecture

Layer	Choice
Framework	Next.js (App Router), frontend and backend in one project
Hosting	Vercel, free Hobby plan (personal project)
API	Route Handlers / Server Actions
Database	MongoDB Atlas, free M0 cluster (512 MB, enough for one event)
ORM	Mongoose (or the official MongoDB Node driver)
Auth	Auth.js (credentials) for staff only (admins and game masters), with role checks on every admin route and action
Real-time updates	Polling (every 3–5 s) with SWR, or a managed service (Pusher / Ably) since Vercel serverless does not keep WebSocket connections
UI	Tailwind CSS, shadcn/ui

Timer synchronization: the server stores `startedAt`; clients compute elapsed time from it, so all devices stay consistent even with refreshes.

9. Data Model (draft)

Each entity below is a MongoDB collection with an ObjectId `_id`, and links between entities store the other document's id. Clues live in their own collection so their content is never returned by public queries.

- **Student**: id, fullName, year (A1/A2), className, gender, mode (TEAM/SOLO), teamId?
- **Team**: id, name, color, clueCount (derived from Clue)
- **Game**: id, name, category (TEAM/BOARD), status (PENDING/RUNNING/DONE), inRotation (boolean, false when frozen), startedAt, endedAt, masterIds (staff assigned to the game; admins can be assigned too)
- **Admin**: id, name, email, passwordHash, role (SUPER_ADMIN, ADMIN or GAME_MASTER)

- **GameAssignment**: id, gameId, teamId, round, status
- **Result**: id, gameId, teamId, isWinner
- **Clue**: id, gameId, teamId?, content, awardedAt
- **EventState**: phase (SETUP, RUNNING, FINAL, DONE), cluesRevealed (boolean)
- **AuditLog**: id, adminId, action, payload, createdAt

10. Non-Functional Requirements

- **Performance:** public pages load in under 2 s on mobile; live data refresh within 5 s.
- **Security:** admin routes protected server-side; clue contents never exposed by public APIs; rate limiting on login.
- **Reliability:** the app must tolerate page refreshes and reconnections during the day; all state lives in the database.
- **Usability:** mobile-first for participants, tablet/phone-friendly for game masters (large Start/End buttons).
- **Accessibility and language:** readable contrast, one language at first (English or French), extensible.

11. Event Flow (day of)

1. **Before:** import the form, generate and lock the teams, create the games, assign game masters, generate the rotation schedule, enter clues.
2. **Round start:** each game master presses Start, and timers begin.
3. **Round end:** the game master presses End, designates the winner, awards the clue to them, and teams rotate to their next game.
4. **Repeat** until every team completed the full rotation.
5. **Final round:** teams gather in a room, the clues are revealed, all teams answer at the same time, and the winner gets the prize.

12. Special Cases and Decisions

1. **Number of teams vs 9 games.** If T is below 9, the super admin freezes games out of the rotation pool (or puts them back). If T is above 9, a team that finishes its rotation of 9 games goes to the board games and keeps playing there; its clues stay, and they are revealed in the app only after every team is done, followed by the final round.

2. **Ties.** The final round is the tie-break. If the tied teams give the same answer, one last game of luck (like a penalty shootout) decides the winner. This is unlikely, since teams rarely end with the same number of clues.
3. **Draw in a game.** The winner is decided by a penalty-style game chosen on the spot, so the clue always goes to one team.
4. **Team or solo.** The form has a field where each student picks Team or Solo.
5. **Team size.** T is the number of team registrations ÷ 6. If it is not a multiple of 6, some teams have 5 or 7 players. Nobody is moved between solo and team against their choice.
6. **Clues and the final answer.** Clues are entered in the app later and shown to the teams in the app at the final. Answers are given orally.
7. **Board games.** They are team vs team, and the winner gets a clue like in the team games (for example Flip 7 or Code Names).
8. **Still open.** The total number of students, which is known only when registration closes.

13. Deliverables and Planning (suggested)

Phase	Content
1	Data model, DB setup, form import, team balancing algorithm
2	Admin auth and dashboard, game start/end, timer
3	Rotation, scoring, clue system
4	Public landing page, solo players page, live refresh
5	Final reveal, tests, dry run, deployment on Vercel

14. Future Extensions

- The 4 board games and their scoring for solo players.
- Photo gallery and announcements on the landing page.
- Export of final results (PDF/CSV).